$$v = \frac{1}{4} v_0 \ . \qquad (9)$$

3. Post-impact A , B , C system CM: position and velocity. Impact is brief, so just after impact, A , B , C have their initial CM position. Write it as $(x_c$, $y_c)$. The definition of the center of mass (CM) gives

$$x_c = \frac{ml\cos\alpha - ml}{3m} \ , \qquad (10)$$

$$y_c = \frac{ml\sin\alpha}{3m} \ . \qquad (11)$$

Substituting the data gives

$$x_c = -\frac{1}{6} l \ , \qquad (12)$$

$$y_c = \frac{\sqrt{3}}{6} l \ . \qquad (13)$$

The post-impact center-of-mass velocity v_c has components

$$v_{cx} = \frac{mv_1 + mv_2\cos\theta - mv_3\cos\alpha}{3m} \ , \qquad (14)$$

$$v_{cy} = \frac{-mv_2\sin\theta - mv_3\sin\alpha}{3m} \ . \qquad (15)$$

Equations (4)-(7), (14), (15) and α give

$$v_{cx} = 0 \ , \qquad (16)$$

$$v_{cy} = -\frac{5}{12} v_0 \ . \qquad (17)$$

4. Motion of A , B , C's center of mass and D. Just after impact, A , B , C's CM starts at $(x_c = -l/6$, $y_c = \sqrt{3}l/6)$, moving toward negative y at a speed of $5\ v_0 / 12$. D starts from origin O toward positive y at speed $v_0 / 4$. Both move uniformly. A , B , C's CM and D move oppositely on parallel lines. If D and C do not collide, v_C , v_D are constant; CM approaches D until y coordinates coincide, when the distance is minimal. Let t be that time. Then

$$vt = y_c + v_{cy}\, t \qquad (18)$$

Substituting v_{cy} , v , y_c gives

$$t = \frac{\sqrt{3}l}{4v_0} \ . \qquad (19)$$

Then D and the center of mass of A , B , C are separated by $l/6$. In deriving (19) We assumed during $t = \sqrt{3}l / 4v_0$, C does not collide with D. This is verified below. Since $v_3 = \sqrt{3}v_0 / 3$, its x component is $\sqrt{3}v_0 / 6$. In t it moves in negative x by